2.12.2 Internet Options

A web browser is a graphical interface between a user and the Internet. Common web browsers include Microsoft's Internet Explorer (commonly called IE), Microsoft's Edge (which is the Windows 10 IE replacement), Mozilla's Firefox, and Google's Chrome. Because Internet Explorer ships with Windows, most textbooks use this browser to explain concepts instead of all the rest.

Most browsers are customizable, and many of the settings relate to security. To get to those settings, click on *Tools* or the Tools icon that looks like a gear in the top-right corner of Internet Explorer (see Figure 2.1). The *Tools* menu option or the gear icon in new versions has configuration items that can be used for web browsing issues. For example, the Compatibility View settings menu option is used when a particular web site doesn't display properly. Internet Explorer version 11 has seven main Internet Options tabs (see Figure 2.2).

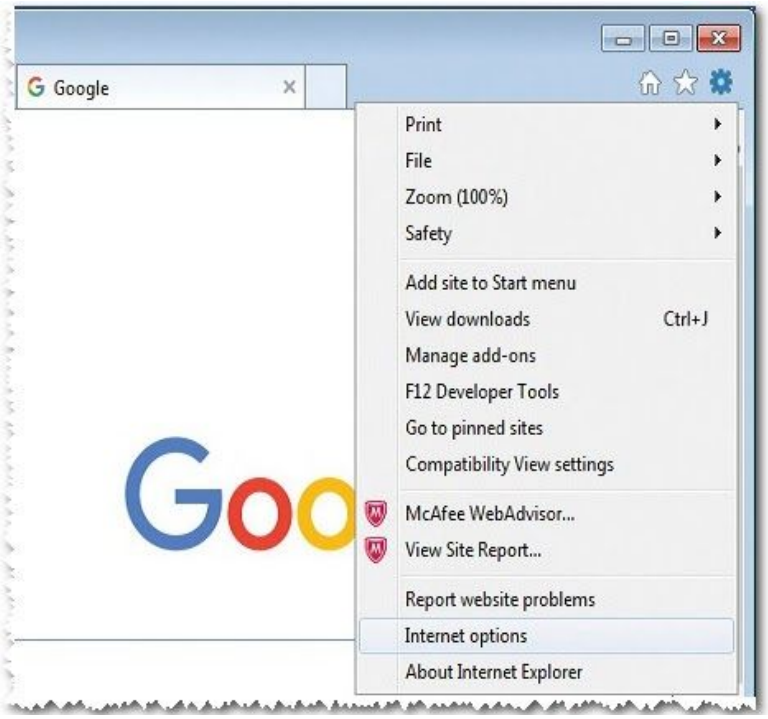


FIGURE 2.1: How to get to Internet Options

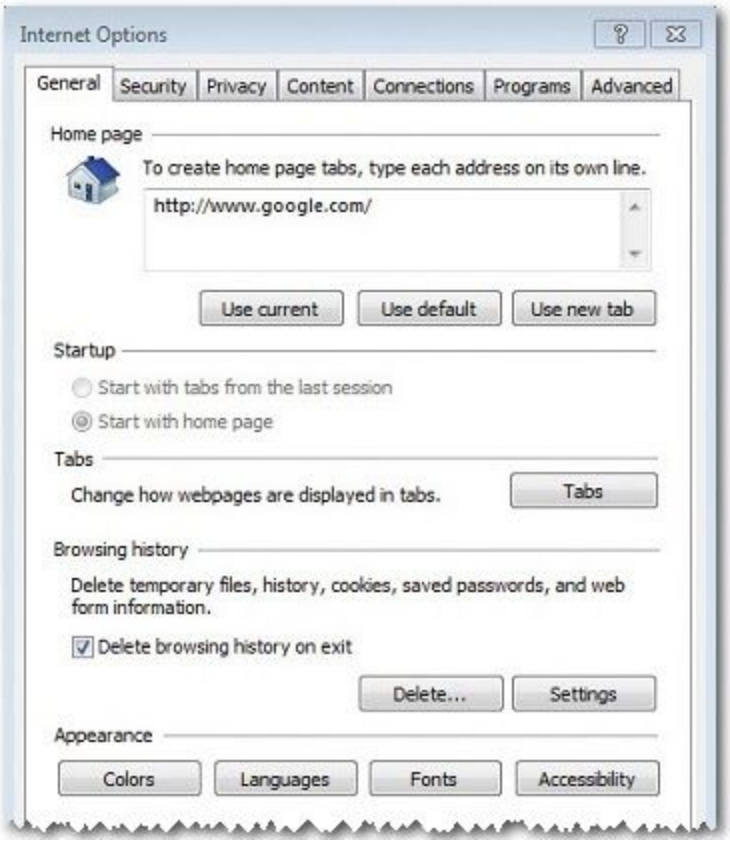


FIGURE 2.2: Internet Options window

A user uses these tabs to configure the browser experience. Note that these options can also be reached using the Internet Options Control Panel in all versions of Windows. Table 2.1 explains the main purposes of the main tabs.

Tab	Purpose
General	Configures the home page (the page that opens every time Internet Explorer opens or the home icon is clicked), deletes or configures how long the browsing history (the websites visited) is kept, configures how tabs are organized and behave, and enables customization of the font, language, and color scheme.
Security	Customizes security options for sites that you trust and ones that you want blocked

Privacy	Configures how your private information is handled, including cookies and pop-ups
Content	Contains Parental Controls, Content Advisor, control security certificates, AutoComplete for ease of completing online forms, and Feeds and Web Slices for providing updated content directly into the browser.
Connections	Configures dial-up (phone line) or VPN (virtual private network) information.
Programs	Configures email access, add-ons such as toolbars and extensions, and HTML-editing program options.
Advanced	Shows a list of options that might be set throughout the other tabs, also presented here in one easily configured list of checkboxes.

TABLE 2.1: Purposes of the Internet Options tabs

Internet Options *General* Tab

The *General* tab is one of the most commonly used tabs. Table 2.2 explains the purpose of its main sections.

Section	Purpose
Home page	Configures the page that opens every time Internet Explorer opens or the home icon is clicked.
Startup	Sets whether to start IE with tabs previously opened or with the home page website.
Tabs	Configures the browsing tabs (not the configuration tabs) and other features, such as warnings and pop-ups (see Figure 2.5).

Browsing history	Deletes or configures how long the browsing history (the websites visited) is kept. If you click on the Settings tab, three tabs are available. The *Temporary Internet Files* tab specifies the maximum amount of space to use (and where that data is stored). See Figure 2.3. The *History* tab sets the number of days to keep a history of your browsing (see Figure 2.6). Use the *Caches and databases* tab to set the amount of cache storage space can be used before displaying a message (see Figure 2.6).
Appearance	Enables you to customize the browser environment, including the colors (see Figure 2.4), language (see Figure 2.4), fonts (see Figure 2.7), and accessibility options (see Figure 2.7).

TABLE 2.2: Internet Explorer General tab sections



FIGURE 2.3: Internet Options General tab > Temporary Internet Files tab

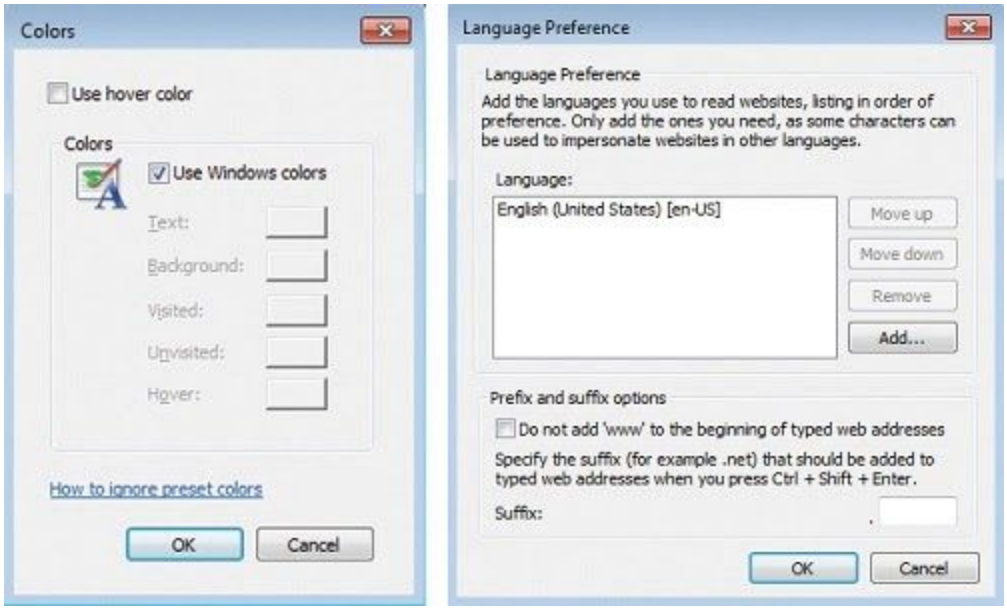


FIGURE 2.4: Internet Options General tab > Colors and Languages windows

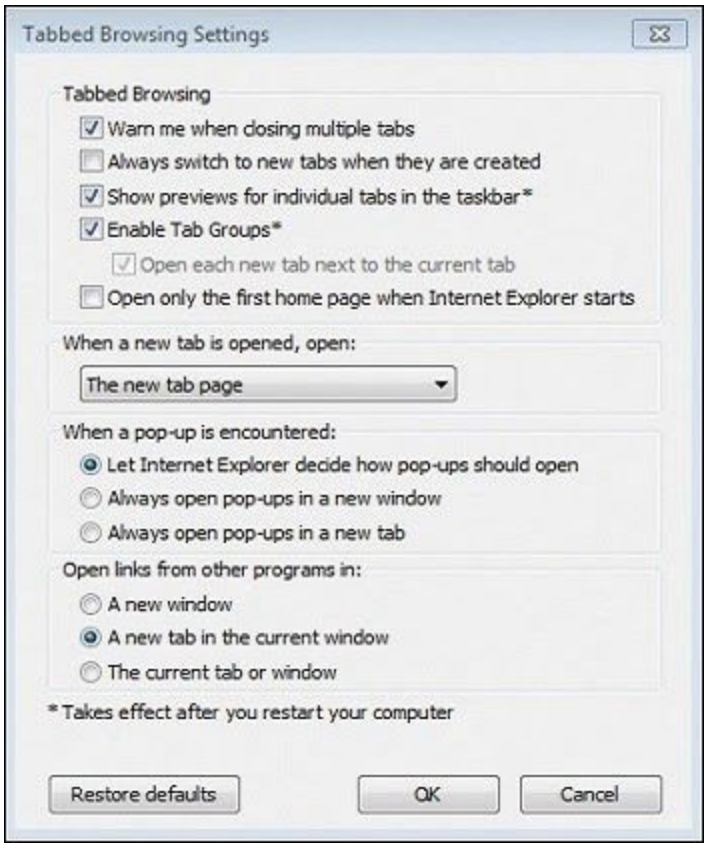


FIGURE 2.5: Internet Options General tab > Tabs button



FIGURE 2.6: Internet Options General tab > History and Caches and databases windows

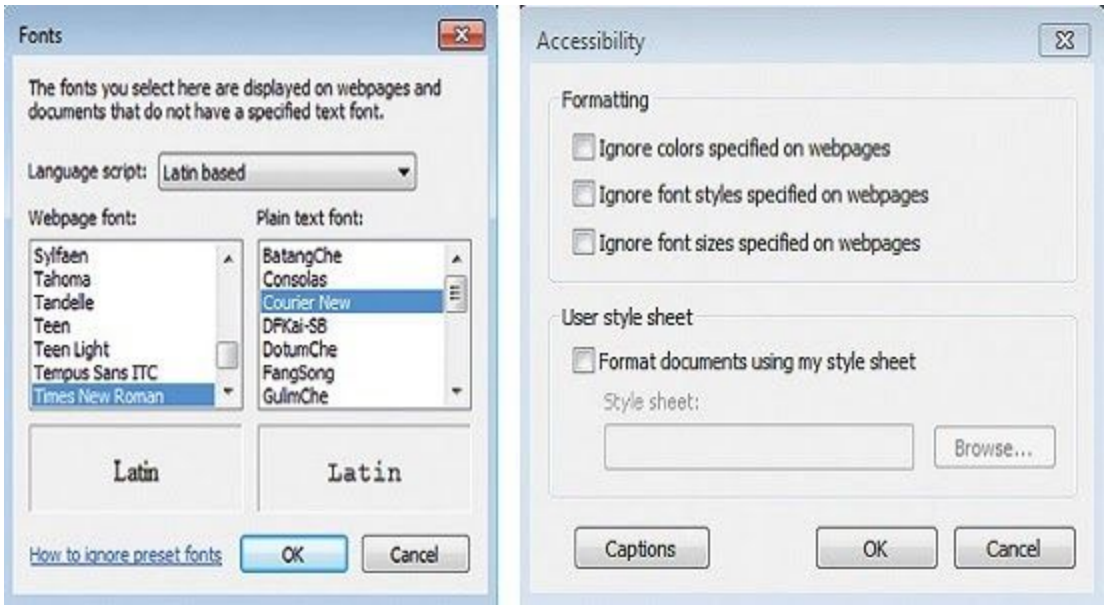


FIGURE 2.7: Internet Options General tab > Fonts and Accessibility windows

Internet Options *Security* Tab

The Internet Options *Security* tab is used to configure security settings related to dangerous or risky online content. Figure 2.8 shows the *Security* tab window. Table 2.3 explains the differences between the four zones.

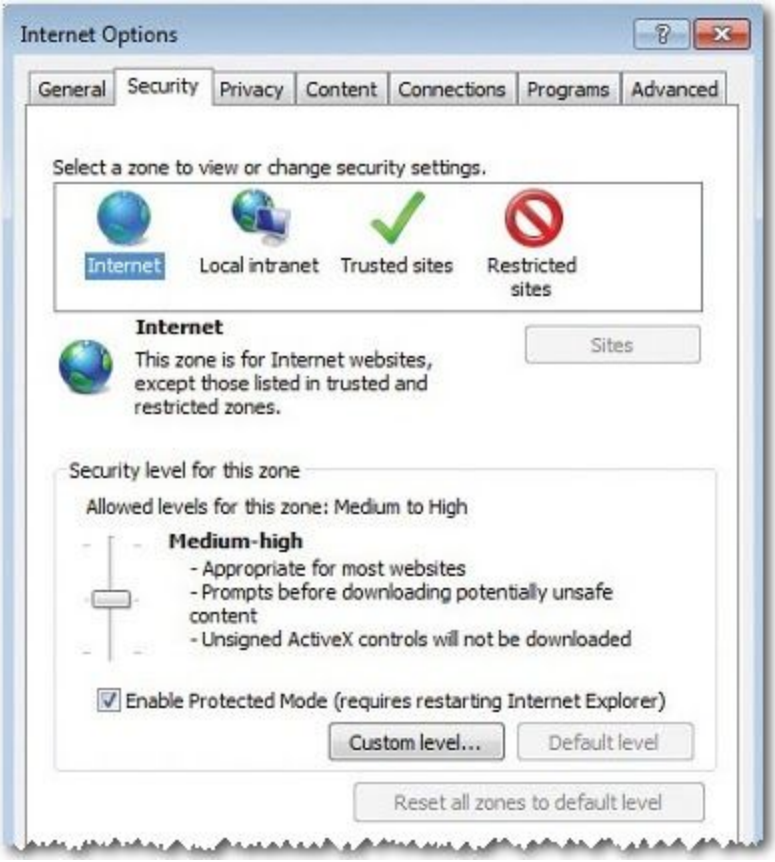


Figure 2.8: Internet Options Security tab

Zone	Purpose
Internet	Sets the security level related to websites (excluding sites from the other three zones)

Local intranet	Sets the security level related to a private network such as the one found inside a company (as opposed to the Internet, which is the rest of the world)
Trusted sites	Sets the security level related to websites you believe do not have content that will infect the computer
Restricted sites	Sets the security level for websites you definitely do not trust

Table 2.3: Internet Explorer Security tab zones

Internet Options *Privacy* Tab

The Internet Options *Privacy* tab is used to configure settings related to your personal information and how it is used by others (see Figure 2.9). Table 2.4 explains the information that can be configured within each section of the *Privacy* tab window.

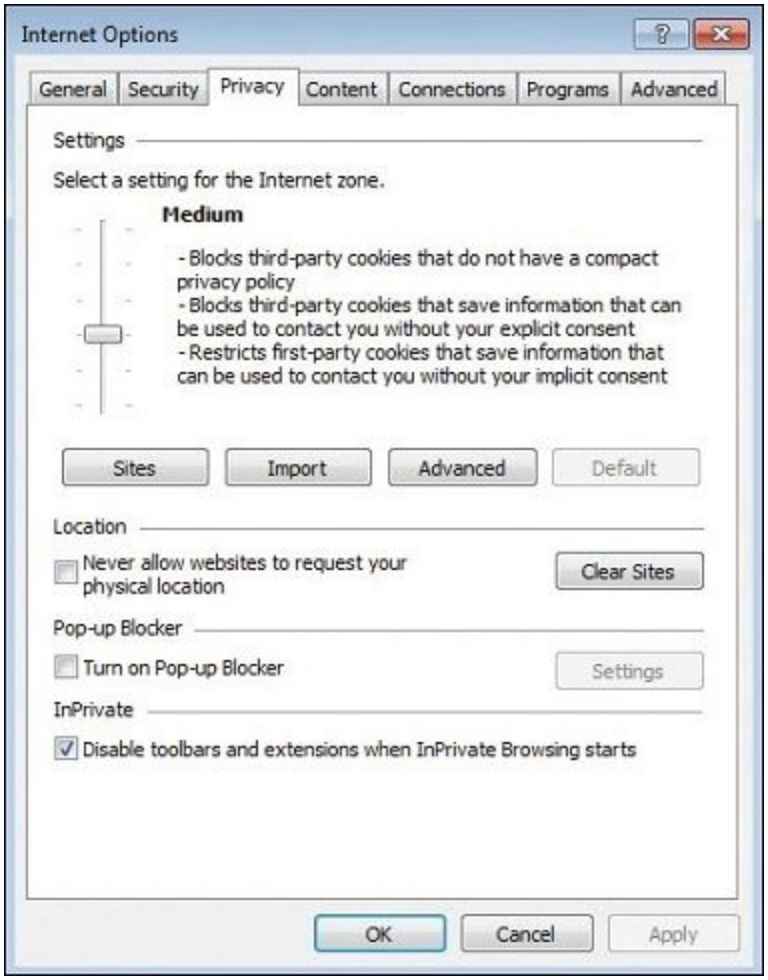


Figure 2.9: Internet Options Privacy tab

Section	Purpose
Settings	Sets the security level for how cookies are handled. The *Sites* button enables you to select whether a site is always allowed or never allowed (blocked) to use cookies (see Figure 2.10). The *Advanced* button enables you to choose how cookies are automatically handled (see Figure 2.10).
Location	Configures whether websites can even request to know where you are located using location services. Even if you allow this, you are prompted whether or not to allow once or always allow the site to know your physical location.

Pop-up Blocker	Controls pop-ups from websites. If you select the option, the *Settings* button is enabled. There, you can select which websites are allowed to produce pop-ups as shown in Figure 2.11.
InPrivate	A configuration that prevents IE/Edge from storing history information and passwords.

Table 2.4: Internet Explorer Privacy tab window

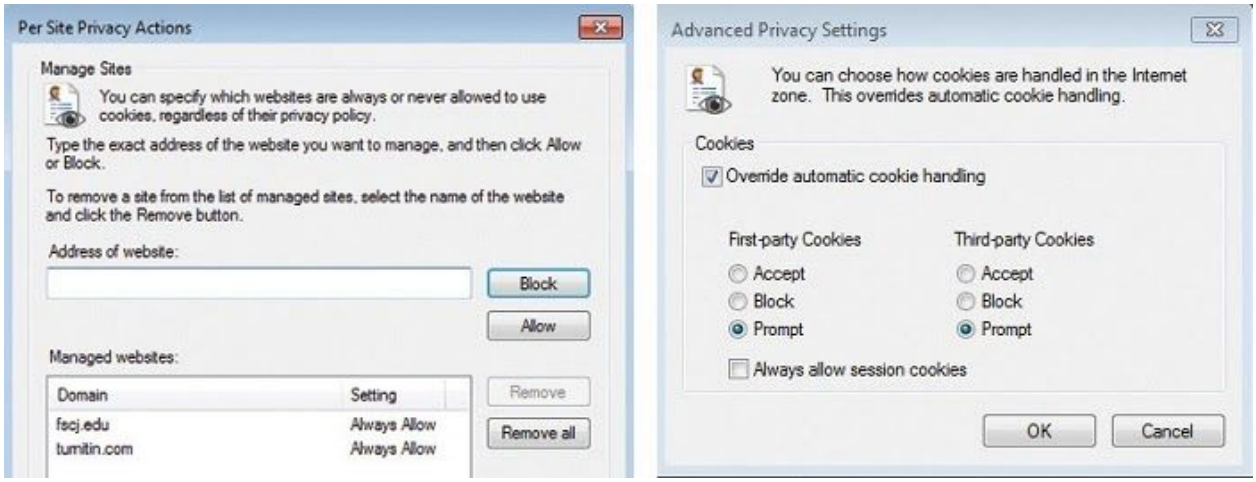


Figure 2.10: Internet Options Privacy tab > Sites and Advanced Privacy Settings windows



Figure 2.11: Internet Options Privacy tab > Pop-up Blocker Settings windows

Private browsing might be important to use in a corporate environment, when using a browser on someone else's computer, going to a website you have never used before, or using a public computer. Here is how to do it in some common browsers:

- Microsoft Internet Explorer: On a desktop, right-click on the *IE/Edge* icon > *Start InPrivate Browsing*. For a mobile device, open *IE* > *Tools* > *Safety* > *InPrivate Browsing.*
- Microsoft Edge: *More actions* (three dots option in top-right corner) > *New InPrivate Window*
- Mozilla Firefox: *Tools* > *Start Private Browsing*
- Google Chrome: Wrench icon > New Incognito window

Internet Options *Content* Tab

The *Content* tab is used to change settings related to what can be seen in the browser window. Figure 2.12 shows the overall screen and Table 2.5 details what each section is for.



Figure 2.12: Internet Options Content tab

Section	Purpose
Family Safety	Links to users and what controls are applied (if any). See Figure 2.13 to see the Parental Controls window and the security warnings that can occur there.
Certificates	Used to verify the authenticity of a person, file, or device provided when doing online purchases or sending an encrypted file.

AutoComplete	The *Settings* button allows settings related to filling in information on a web page such as your name, address, and email address (see Figure 2.14 left).
Feeds and Web Slices	Controls settings related to rich site summary (RSS) feeds to receive data, audio, and video information from news sources and the like (see Figure 2.14 right)

Table 2.5: Internet Explorer Content tab window



Figure 2.13: Internet Options Content tab > Parental Controls window

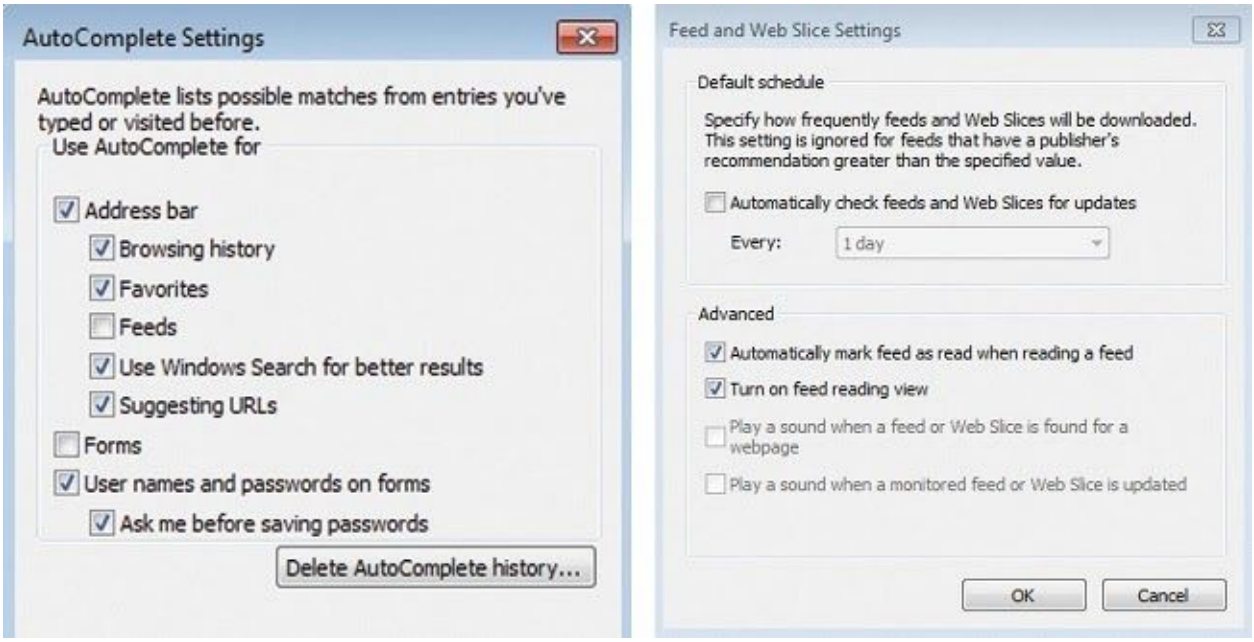


Figure 2.14: Internet Options Content tab > AutoComplete Settings window and Feeds and Web Slices Settings window

Internet Options *Connections* Tab

The *Connections* tab (see Figure 2.15) is where you go to configure your Internet connection for a newly installed type (see Figure 2.17), set up a new VPN so that someone can securely access company resources while working from home or away from the office (see Figure 2.16), and connections related to your LAN settings (see Figure 2.18).



Figure 2.15: Internet Options Connections tab



Figure 2.16: Internet Options Connections tab > Add VPN connection window

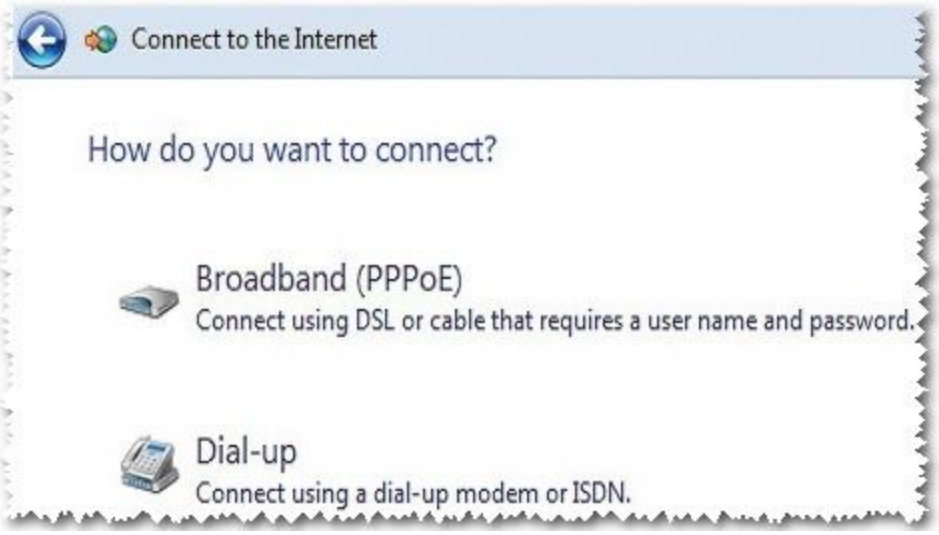


Figure 2.17: Internet Options Connections tab > Setup Internet connection window

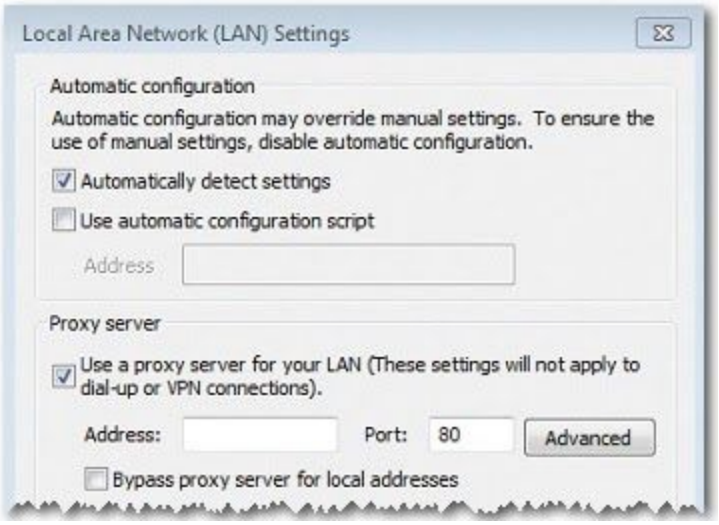


Figure 2.18: Internet Options Connections tab > LAN settings window

Internet Options *Programs* Tab

How many times do you think someone downloads a software update of some type and gets the default browser changed to something else? The Programs tab (see Figure 2.19) can help with that and with managing add-ons. Add-ons are extensions or plug-ins that provide the browser an additional feature such as a toolbar or the capability to dim everything on the screen except for a

running video. Add-ons can also cause security risks. Figure 2.20 shows the Manage Add-ons window.

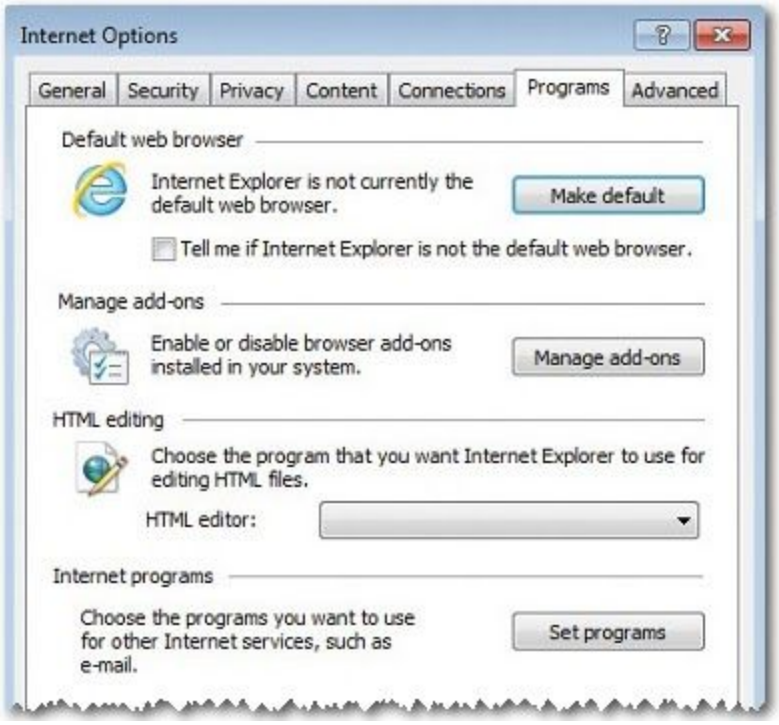


Figure 2.19: Internet Options Programs tab

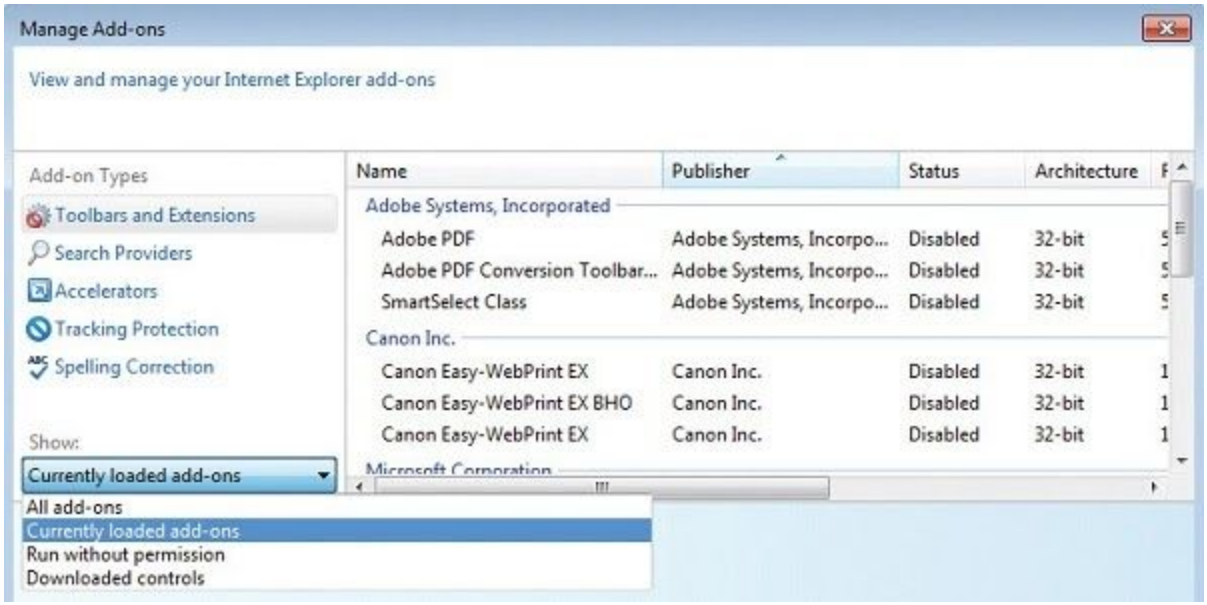


Figure 2.20: Internet Options Programs tab > Manage add-ons window

The HTML editing section has a drop-down menu to select the application used to edit HTML files. Examples include Microsoft Windows, Excel, and Notepad. The last section is used to access the Set Default Programs Control Panel where you can select which application opens a music file, for example. It is similar to choosing your default browser, but for other applications.

Internet Options *Advanced* Tab

The last tab in *Internet Options* is the *Advanced* tab (see Figure 2.21). This tab is used for all types of options not covered on the other tabs. Table 2.6 outlines just a few examples of the sections.

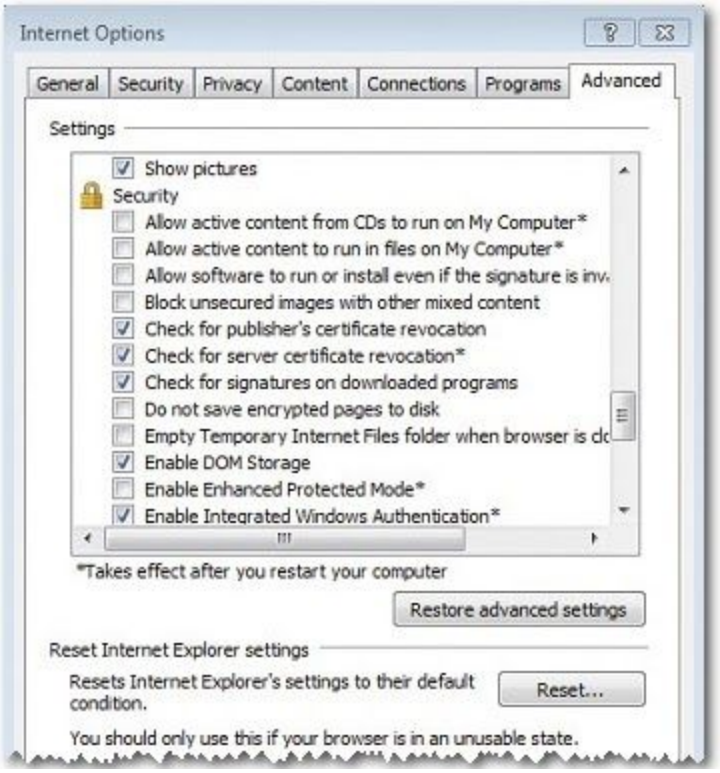


Figure 2.21: Internet Options Advanced tab

Section	Purpose

Accelerated graphics	Used to enable software rendering instead of GPU rendering
Accessibility	Used to play system sounds, provide ALT text for images, and reset text sizes
Browsing	Used to set browsing options such as notification of script errors or when a download completes, automatic crash recovery, and loading content in the background to improve performance
HTTP settings	Used to set the HTTP version
International	Used to set settings such as Unicode UTF-8 format, which is a standard way to display characters found in different languages
Multimedia	Used to set whether animations and sounds play
Security	Used to select which version(s) of secure sockets layer (SSL) and transport layer security (TLS) are enabled

Table 2.6: Internet Explorer Advanced tab window